

Project KRAKEN

Central European Olympiad in Artificial Intelligence

Preamble: The Event Horizon Paradox

“We thought the void was silent. We were wrong. It is not that the universe does not speak; it is that our ears were never designed to hear the frequency of its screaming.”

— *Dr. Aris Thorne, Lead Astrometric Specialist, DSIA*

The Context

At the edge of the known sector lies X-99, a Supermassive Singularity that defies the standard model of cosmic evolution. It is not merely a gravity well; it is a wound in the skin of reality.

Last week, the patterns changed.

The singularity has entered an unstable resonance phase. The event horizon is no longer static but a shifting membrane. We are detecting high-dimensional geometric anomalies — “Leviathan-Class” hyper-entities.

They exist natively in $\mathbb{R}^5$, but as they intersect our four-dimensional brane, they cast gravitational shadows that distort time and space.

The Challenge

You are the architect of the **KRAKEN Protocol**, an AI system designed to interpret these signals. The data is warped by relativistic dilation and corrupted by Hawking radiation noise.

Only machine learning can reconstruct the shape of what rises from the abyss.

1 Mathematical Definition

The data exists on a pseudo-Riemannian manifold $\mathcal{M}$ with metric tensor $g_{\mu\nu}$. Let $\mathbf{X} \in \mathbb{R}^5$ denote the latent state. The entity undergoes a transformation $\mathcal{T} \in SO(1, 4)$.

The observed signal is:

$$S_{\text{obs}}(\tau) = \int_0^{\Delta\tau} \sqrt{-g} \left[\mathcal{F}(\Psi(x^\mu)) e^{-i\omega t} + \sum_{n=0}^{\infty} \frac{1}{n!} \mathcal{L}_{\text{interaction}}^{(n)} \right] dt + \eta(t)$$

Where:

- $g = \det(g_{\mu\nu})$
- $\mathcal{F}$ is the Fourier transform
- $\eta(t) \sim \mathcal{N}(0, \sigma^2(t))$

Objective: Reverse-engineer the diffeomorphism to classify topology and predict collapse.

2 Dataset Specification

WARNING: Raw interferometric data.

2.1 1. Holographic Biopsies (Slices)

- Files: `train_slices.npy`, `test_slices.npy`
- Dimensions: $(N, 3, 128, 128)$
- Channels:
 1. Real ($\Re$): Gravimetric potential
 2. Imaginary ($\Im$): Frame-dragging momentum
 3. Spin-tensor (σ): Vorticity field

2.2 2. Echoes

- Files: `train_echoes.npy`, `test_echoes.npy`
- Dimensions: $(N, 1024, 2)$
- Signal: $h(t) = [h_+, h_-]$
- Corruption: Quantum gaps

2.3 3. Glyphs

- Files: `train_glyphs.csv`, `test_glyphs.csv`
- Grammar depends on eigenvalues of channel 0

2.4 4. Ground Truth

`train_targets.csv`

3 The Mission

3.1 Subtask 1: Geodesic Rectification (30 points)

Predict spline coefficients $\mathbf{w} \in \mathbb{R}^{10}$ of warping function $\Omega(t)$.

Metric: Mean Squared Error (MSE)

$$\text{thres}_{\max} = 0.00085, \quad \text{thres}_{\min} = 0.020$$

$$s_1 = \begin{cases} 30 & \text{if } \text{MSE} \leq 0.00085 \\ 0 & \text{if } \text{MSE} \geq 0.020 \\ \left\lfloor 30 \cdot \frac{0.020 - \text{MSE}}{0.020 - 0.00085} \right\rfloor & \text{otherwise} \end{cases}$$

3.2 Subtask 2: Topological Classification (35 points)

Predict class $y \in \{0, \dots, 49\}$ or -1 .

Metric: Macro F1 (penalized)

$$\text{thres}_{\max} = 0.85, \quad \text{thres}_{\min} = 0.55$$

$$s_2 = \begin{cases} 35 & \text{if } F1 \geq 0.85 \\ 0 & \text{if } F1 \leq 0.55 \\ \left\lfloor 35 \cdot \frac{F1 - 0.55}{0.85 - 0.55} \right\rfloor & \text{otherwise} \end{cases}$$

3.3 Subtask 3: Stability Prediction (35 points)

Predict $s \in [0, 1]$.

Metric: Root Mean Squared Error (RMSE)

$$\text{thres}_{\max} = 0.35, \quad \text{thres}_{\min} = 0.50$$

$$s_3 = \begin{cases} 35 & \text{if } \text{RMSE} \leq 0.35 \\ 0 & \text{if } \text{RMSE} \geq 0.50 \\ \left\lfloor 35 \cdot \frac{0.50 - \text{RMSE}}{0.50 - 0.35} \right\rfloor & \text{otherwise} \end{cases}$$

Final Score

$$\text{Total} = s_1 + s_2 + s_3$$

Maximum: **100 points**

4 Submission Format

```
subtaskID,datapointID,answer
1,test_00000,"0.123;0.456;...;0.999"
2,test_00000,-1
3,test_00000,0.884
```

Requirements:

- Subtask 1: 10 floats separated by semicolons inside quotes
- Subtask 2: Integer (0–49 or -1)
- Subtask 3: Float in $[0, 1]$

5 Appendix A: Macro F1 with Penalty

For each class c :

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP'_c}$$

$$FP'_c = \begin{cases} 2 \cdot FP_c & \text{if } c = -1 \\ FP_c & \text{otherwise} \end{cases}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

$$\text{Macro F1} = \frac{1}{|C|} \sum_{c \in C} F1_c$$

6 Appendix B: Technical Context

Warping relation:

$$\beta = \frac{v}{c}, \quad t' = \frac{t - vx}{\sqrt{1 - \beta^2}}$$

Spline coefficients parameterize $\beta(t)$.

Glyph rules:

- ∇ : differentiate next token
- $\int$: integrate next token
- Φ : if $\bar{\phi} > \frac{\pi}{2}$ parse in reverse